

EGR 345 Learning Outcomes for Exam 1

Chapter 2: Laplace

- using Laplace to solve differential equations
- finding transfer functions
- impulse and step response
- partial fraction expansion
- pole locations

Chapter 3: Dynamics of Mechanical Systems

- FBDs
- Mass/Spring/Damper systems
- Equations of motion
- definitions:
 - ζ
 - b_{cr}
 - ω_n
 - ω_d
- effects of ζ and ω_n on step response
- multiple degree of freedom MSD systems
- pole locations for MSD systems